

Group 3 Final Project
An Empirical Study of Book-to-Market Portfolios in the U.S. Market (1982–2019)
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Kaylee Xiaojing Zhang 12487310

Sisly Lyu 72710643

Xiaoxi Xu 12491015

Romee Sun 99320285

Zhuji Zhang 65457343

Wenqiang Yu 71090450

1. Introduction

1.1 Motivation

The book-to-market (B/M) ratio has long been a fundamental metric in asset pricing, serving as a key tool for distinguishing between value and growth stocks. Historically, firms with high B/M ratios—commonly referred to as value stocks—have consistently outperformed their low B/M (growth) counterparts, a phenomenon widely documented in empirical finance. This “value premium” has remained a central theme in both academic research and practical investment strategies.

However, recent literature has raised questions about the consistency and economic drivers of this premium. Notably, studies such as Hou, Xue, and Zhang (2020) and Fama & French (2015) highlight that the explanatory power of B/M may have declined in recent decades, and that factors like profitability and investment may offer better explanatory power.

Motivated by this, we conduct additional analyses including subperiod regressions (pre-2000 vs. post-2000), double sorting by size and B/M, and re-estimation using the Fama-French five-factor model to test the robustness and persistence of the value effect.

1.2 Literature Review

Fama and French (1992) were among the first to formally demonstrate the explanatory power of the B/M ratio in cross-sectional stock returns, subsequently incorporating it into their influential three-factor model (Fama and French, 1993). This model augmented the traditional CAPM by introducing size (SMB) and value (HML) factors, providing a significantly improved fit to empirical return patterns.

Recognizing the limitations of their original model, Fama and French (2015) later extended the framework to a five-factor specification by adding profitability (RMW) and investment (CMA) factors. While the new model improved overall fit, it also reduced the statistical significance of the value factor (HML), raising questions about the continued robustness of the B/M effect.

More recently, Hou, Xue, and Zhang (2020) conducted a large-scale replication of cross-sectional anomalies and found that the predictive power of the B/M ratio significantly declined post-2000. Their findings motivate our decision to split the sample and re-evaluate the value premium in pre- and post-2000 subperiods.

Furthermore, Lakonishok, Shleifer, and Vishny (1994) offered an alternative behavioral explanation for the value premium, arguing that it may result from investor biases and

mispricing, rather than compensation for risk—an idea we revisit through our analysis of long-short abnormal returns.

1.3 Contribution and Key Takeaways

In this project, we revisit the book-to-market anomaly using a comprehensive and data-driven approach. We build B/M-sorted portfolios from scratch using CRSP and Compustat data and evaluate performance through CAPM, Fama-French three-factor (FF3), and five-factor (FF5) models. Our investigation includes:

- A time-split analysis of the value premium before and after 2000,
- A comparison of equal-weighted vs. value-weighted returns,
- An extension through double sorting on size and B/M.

Our findings provide new insight into the conditional nature of the value premium and how it has evolved over time. They also underscore the importance of incorporating both firm characteristics and model complexity in evaluating return patterns.

2. Data and Methods

2.1 Data Sources and Sample Construction

Our analysis relies on monthly data from CRSP and Compustat spanning 1982 to 2019. CRSP provides stock-level data on returns, prices, share counts, and market capitalization, while Compustat supplies the necessary accounting variables to construct book equity. We restrict our sample to common stocks listed on NYSE, AMEX, and NASDAQ (share codes 10 and 11; exchange codes 1, 2, and 3).

To mitigate the influence of extreme events, we exclude major crisis years: 1987, 2000–2002, 2008–2009, and 2020. This allows for a cleaner assessment of typical performance across economic cycles.

We construct book equity using the hierarchical fallback logic of Hou, Xue, and Zhang (2020). Specifically, BE is computed as shareholders' equity (SEQ), or CEQ + PSTK if SEQ is missing, or AT - LT otherwise. Deferred taxes (TXDITC) are added when available, and preferred stock (PSTK) is subtracted. We drop observations with $BE \leq 0$ or missing inputs. Book-to-market (B/M) ratio is calculated by dividing BE at fiscal year $t-1$ by market capitalization in June of year t .

2.2 Portfolio Construction

Following Fama and French (1992, 1993), we sort stocks into five portfolios based on B/M ratio each June. These portfolios are held from July of year t to June of year $t+1$. We implement both equal-weighted and value-weighted schemes to account for firm size and influence.

For value-weighting, returns are weighted using each stock's June market capitalization. For equal-weighting, each stock receives equal weight regardless of size. Portfolios are rebalanced annually. For each construction, we calculate monthly returns and form a long-short portfolio by taking a long position in the highest B/M quintile (Q5) and a short position in the lowest quintile (Q1).

Descriptive statistics such as mean return, standard deviation, skewness, and kurtosis are reported for each portfolio. We also conduct t-tests on portfolio means to assess statistical significance.

2.3 Asset Pricing Models

We evaluate portfolio performance using the Capital Asset Pricing Model (CAPM) and the Fama-French three-factor (FF3) model. Regressions are conducted at the monthly level for each quintile portfolio and the long-short portfolio, using excess returns over the risk-free rate as the dependent variable.

Factor data, including the market risk premium (Mkt-RF), size (SMB), and value (HML) factors, are sourced from Ken French's online data library. Alphas (intercepts), factor loadings (betas), t-statistics, p-values, and R-squared values are reported.

2.4 Extension: Five-Factor Model and Double Sorting

While the Fama-French three-factor (FF3) model captures size, market, and value effects, recent academic research suggests that these three factors may not fully explain cross-sectional differences in stock returns.

In particular, Fama and French (2015) argue that the B/M ratio alone may not be sufficient, as value stocks often exhibit high profitability and conservative investment behavior — characteristics that are not accounted for in the FF3 model.

Therefore, we adopt the Fama-French five-factor model, which adds profitability (RMW) and investment (CMA) factors to the original framework. RMW captures differences in firm profitability, while CMA reflects investment behavior.

These factors reflect traits often embedded in value stocks, like strong profitability and prudent investment. By incorporating RMW and CMA, the FF5 model allows us to test whether the observed value premium is distinct or merely a proxy for these underlying traits. If alpha becomes insignificant in the FF5 regressions, this would suggest that B/M alone may not have independent explanatory power.

To further assess the robustness of the value premium, we implement double sorting.

We conduct a double-sorting analysis by first dividing firms into small and big groups based on the median June market capitalization. Within each size group, firms are sorted into five B/M portfolios, resulting in ten double-sorted portfolios (S0-S4 and B0-B4).

We also compute both equal- and value-weighted returns for these portfolios and run CAPM, FF3, and FF5 regressions. This approach helps identify whether the value effect is concentrated among small-cap stocks and whether it persists after controlling for additional risk factors.

All portfolio formation and regressions are conducted in Python, based on enhanced versions of the instructor-provided code. Our modifications support flexible sorting schemes (including double sorting), alternate weighting approaches, crisis-year exclusion, and integration of FF5 model factors.

3. Results and Interpretation

3.1 Descriptive Portfolio Statistics

Table 1 presents monthly return statistics for the B/M-sorted quintile portfolios under both equal-weighted and value-weighted schemes. For equal-weighted portfolios, average returns increase slightly across quintiles, ranging from 1.89% (Q1) to 1.93% (Q5), though the long-short spread (Q5–Q1) is only 0.04% and statistically insignificant ($t = 0.12$).

In contrast, value-weighted portfolios exhibit a clearer monotonic pattern, with Q5 generating a monthly return of 1.68% versus 0.97% for Q1. The resulting long-short return of 0.72% is highly significant ($t = 3.997$), indicating a robust value premium among large-cap stocks.

3.2 CAPM Regression

Table 2 reports CAPM regression results. In the equal-weighted specification, all five portfolios display significant positive alphas at the 1% level, with Q5 showing the highest monthly alpha (0.74%). However, the long-short portfolio yields a slightly negative alpha (–0.05%), suggesting that the CAPM does not capture the return spread across B/M quintiles.

For value-weighted portfolios, Q1 has a significant negative alpha (–0.59%), while Q5 has a significant positive alpha (0.49%), leading to a long-short alpha of 0.77% ($t = 4.52$), which is both statistically and economically meaningful.

3.3 Fama-French Three-Factor (FF3) Model

Table 3 shows FF3 regression outcomes. For equal-weighted portfolios, the inclusion of SMB and HML factors improves model fit (average R^2 increases to 0.91), but over-corrects for the value premium. The long-short alpha becomes -0.41% and is significantly negative ($t = -3.89$), indicating a reversal of the value premium among small-cap stocks.

In the value-weighted case, FF3 reduces alphas modestly but maintains significance. The long-short alpha is 0.50% ($t = 4.72$), confirming that the value effect persists even after accounting for size and value risk factors.

3.4 Model Comparison

In both weighting methods, FF3 consistently achieves higher R^2 values than CAPM, confirming its improved explanatory power. This is especially evident in the equal-weighted case, where R^2 rises from ~ 0.84 to ~ 0.91 . For value-weighted portfolios, the average R^2 increases from ~ 0.66 (CAPM) to ~ 0.87 (FF3).

4. Robustness Analysis

To examine the time-varying nature of the value premium, we divide the sample into two subperiods: 1982–1999 (pre-2000) and 2000–2019 (post-2000), consistent with Hou, Xue, and Zhang (2020).

4.1 Value-Weighted Portfolios

Table 4 shows that the value premium is evident in both periods but has weakened over time.

- **Pre-2000:** The long-short CAPM alpha is 12.68% annually ($t = 4.13$), while FF3 yields 6.41% ($t = 3.29$).
- **Post-2000:** CAPM alpha drops to 5.69% and FF3 to 5.35% , though both remain statistically significant.

4.2 Equal-Weighted Portfolios

Equal-weighted results show limited support for the value premium.

- **Pre-2000:** Long-short alpha is -0.0152 (CAPM) and -10.13% annually (FF3), both insignificant or negative.
- **Post-2000:** Long-short FF3 alpha remains negative (-0.005), implying no value premium among smaller firms.

4.3 Interpretation

These findings suggest that:

1. The value premium has declined in the post-2000 era.
2. It remains stronger among large-cap firms (value-weighted), but disappears or reverses among small caps (equal-weighted).

This reinforces that portfolio construction methods and sample periods substantially affect the observed B/M effect.

5. Extension: Fama-French Five-Factor Model and Double Sorting

5.1 Fama-French Five-Factor (FF5) Model

In Table 5.1, we estimate the FF5 model to test if the value premium persists after controlling for profitability (RMW) and investment (CMA).

- **Equal-Weighted Portfolios:** All five quintiles show significant alphas (annualized: 2.67%–7.42%), with high R^2 (~ 0.92).
- **Value-Weighted Portfolios:** Only Q0, Q1, and Q4 display significant alphas; Q2–Q3 are mostly insignificant, showing a weaker and less monotonic value effect.

5.2 Double Sorting: Size $\times$ B/M (Equal-Weighted)

Table 5.2 presents the results of the 2×5 double sorting by Size and Book-to-Market ratio under both equal-weighted and value-weighted portfolio constructions.

A 2×5 sort is performed each year by first splitting firms by median size, then sorting within each group by B/M into five portfolios.

- **Small Firms (S0–S4):** FF5 alphas increase from -0.46% (S0) to 0.64% (S4), all statistically significant. S4's annual alpha is 7.67% , suggesting a strong value premium in small caps.
- **Large Firms (B0–B4):** FF5 alphas are mostly negative and insignificant, consistent with diminishing value effects in large-cap stocks.

5.3 Double Sorting: Size $\times$ B/M (Value-Weighted)

- **Small Firms (S0–S4):** FF5 alphas remain positive but lower than in equal-weighted portfolios. S4 achieves a 2.96% annual alpha ($t = 15.3$).
- **Large Firms (B0–B4):** Alphas are negligible (0.003–0.007/month) and largely insignificant, reaffirming the absence of a value premium among big firms.

5.4 Interpretation

The double sorting results affirm that the B/M effect is strongest among small firms and weakens significantly in large-cap stocks. Even after controlling for five factors, abnormal returns persist in small-cap high B/M portfolios, confirming the conditional and size-dependent nature of the value anomaly.

6. Reference

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7. Appendix

Table 1

Equal-Weighted Summary:

Metric	Qnt 1	Qnt 2	Qnt 3	Qnt 4	Qnt 5	Long-Short
Mean	0.018909	0.017222	0.017064	0.016330	0.019311	0.000402
t-stat	7.847049	8.327219	8.647225	8.112714	8.921784	0.210046
p-value	0.0	0.0	0.0	0.0	0.0	0.833764

SD	0.043375	0.037226	0.035521	0.036233	0.038961	0.034468
Skewness	0.127295	0.085863	0.042551	-0.071621	0.002408	0.51939
Kurtosis	0.685151	0.882168	0.702045	1.058057	0.929761	1.829796

Value-Weighted Summary:

Metric	Qnt 1	Qnt 2	Qnt 3	Qnt 4	Qnt 5	Long-Short
Mean	0.009661	0.012206	0.013388	0.014315	0.01682	0.007159
t-stat	3.047329	4.524996	5.467639	6.350961	6.427642	3.996705
p-value	0.002517	0.000009	0.0	0.0	0.0	0.000081
SD	0.054638	0.046486	0.042199	0.038844	0.045098	0.030869
Skewness	0.13183	-0.054007	-0.114582	-0.079219	0.346366	-0.210458
Kurtosis	0.938422	0.778502	0.849817	0.874021	2.43344	1.213641

Table 2

Equal-Weighted CAPM:

Metric	Qnt 1	Qnt 2	Qnt 3	Qnt 4	Qnt 5	Long-Short
Alpha	0.004886	0.004414	0.004864	0.004491	0.007439	-0.000537
Alpha t	5.15465	7.147215	7.09652	4.654099	5.889483	-0.276028
Alpha p	0.0	0.0	0.0	0.000005	0.0	0.782704
Beta Mkt	1.126486	1.001737	0.939336	0.902267	0.905676	-0.216439
Beta Mkt t	43.881314	59.888699	50.59802	34.518541	26.472559	-4.108962
Beta Mkt p	0.0	0.0	0.0	0.0	0.0	0.00005
R²	0.856734	0.917619	0.888278	0.787252	0.685177	0.049821

Value-Weighted CAPM:

Metric	Qnt 1	Qnt 2	Qnt 3	Qnt 4	Qnt 5	Long-Short
Alpha	-0.005869	-0.002336	0.000082	0.002021	0.004897	0.007692
Alpha t	-2.943795	-1.656139	0.06131	1.550015	2.46653	4.524945
Alpha p	0.0035	0.098757	0.951153	0.12221	0.014211	0.000009

Beta Mkt	1.236206	1.138293	1.015904	0.915685	0.87901	-0.354527
Beta Mkt t	22.844677	29.733987	27.852664	25.871139	16.312712	-7.68375
Beta Mkt p	0.0	0.0	0.0	0.0	0.0	0.0
R²	0.63887	0.749811	0.724498	0.694084	0.474251	0.166761

Table 3

Equal-Weighted FF3:

Metric	Qnt 1	Qnt 2	Qnt 3	Qnt 4	Qnt 5	Long-Short
Alpha	0.006432	0.004301	0.004054	0.002706	0.005437	-0.004067
Alpha t	9.822696	7.234052	6.800706	4.453202	6.581623	-3.888273
Alpha p	0.0	0.0	0.0	0.000012	0.0	0.000123
Beta Mkt	1.03889	1.014342	0.991234	1.010124	1.013944	-0.021952
Beta Mkt t	56.513347	60.777565	59.226211	59.210678	43.718171	-0.747617
Beta Mkt p	0.0	0.0	0.0	0.0	0.0	0.45524
Beta SMB	-0.092434	-0.119587	-0.074803	-0.030908	0.223627	0.321846
Beta SMB t	-4.029174	-5.741768	-3.581449	-1.451775	7.726375	8.783415
Beta SMB p	0.00007	0.0	0.000395	0.147544	0.0	0.0
Beta HML	-0.526686	-0.032968	0.206006	0.530069	0.740502	1.26484
Beta HML t	-18.943174	-1.306072	8.138367	20.543622	21.110271	28.481749
Beta HML p	0.0	0.192466	0.0	0.0	0.0	0.0
R²	0.933201	0.925428	0.91744	0.9176	0.868448	0.731554

Value-Weighted FF3:

Metric	Qnt 1	Qnt 2	Qnt 3	Qnt 4	Qnt 5	Long-Short
Alpha	-0.004318	-0.001936	-0.000259	0.001319	0.003893	0.005147
Alpha t	-3.794329	-2.276783	-0.326337	1.814097	2.930972	4.721050
Alpha p	0.000180	0.023520	0.744403	0.070686	0.003645	0.000004
Beta Mkt	1.071119	1.057607	0.980733	0.904422	0.866566	-0.203024
Beta Mkt t	33.702193	44.549094	44.315998	44.534582	23.360271	-6.668092

Beta Mkt p	0.0	0.0	0.0	0.0	0.0	0.0
Beta SMB	0.846876	0.667665	0.656873	0.631045	0.858857	0.018740
Beta SMB t	21.220661	22.397124	23.637987	24.746041	18.438081	0.490152
Beta SMB p	0.0	0.0	0.0	0.0	0.0	0.624393
Beta HML	-0.213691	0.118985	0.373972	0.490162	0.683884	0.896895
Beta HML t	-4.458999	3.323839	11.206767	16.006572	12.226178	19.535597
Beta HML p	0.000012	0.001001	0.0	0.0	0.0	0.0
R²	0.884597	0.910866	0.906041	0.906697	0.769168	0.663862

Table 4

Value-Weighted CAPM (Pre-2000):

	Alpha	Alpha_t	Alpha_p	R²	Alpha_Annual
Qnt 1	-0.0099	-3.2018	0.0017	0.5905	-0.1188
Qnt 2	-0.0039	-1.7176	0.0880	0.6875	-0.0465
Qnt 3	-0.0002	-0.0960	0.9237	0.6552	-0.0024
Qnt 4	0.0016	0.8799	0.3803	0.6591	0.0194
Qnt 5	0.0056	1.8826	0.0618	0.3629	0.0674

Value-Weighted CAPM (Post-2000):

	Alpha	Alpha_t	Alpha_p	R²	Alpha_Annual
Qnt 1	-0.0019	-0.7901	0.4307	0.7068	-0.0234
Qnt 2	-0.0009	-0.5441	0.5872	0.8262	-0.0108
Qnt 3	0.0002	0.1490	0.8818	0.8062	0.0029
Qnt 4	0.0023	1.2900	0.1991	0.7427	0.0277
Qnt 5	0.0040	1.5854	0.1150	0.6056	0.0481

Value-Weighted FF3 (Pre-2000):

	Alpha	Alpha_t	Alpha_p	R²	Alpha_Annual
Qnt 1	-0.0067	-3.9957	0.0001	0.8854	-0.0807

Qnt 2	-0.0024	-2.0607	0.0411	0.9239	-0.0283
Qnt 3	-0.0002	-0.1882	0.8516	0.8915	-0.0027
Qnt 4	0.0006	0.5192	0.6044	0.8838	0.0069
Qnt 5	0.0036	1.8961	0.0600	0.7615	0.0428

Value-Weighted FF3 (Post-2000):

	Alpha	Alpha_t	Alpha_p	R²	Alpha_Annual
Qnt 1	-0.0016	-1.2085	0.2288	0.9189	-0.0189
Qnt 2	-0.0007	-0.7655	0.4452	0.9404	-0.0089
Qnt 3	0.0003	0.3768	0.7069	0.9486	0.0038
Qnt 4	0.0024	2.5946	0.0104	0.9341	0.0284
Qnt 5	0.0041	2.2897	0.0235	0.8052	0.0491

Equal-Weighted CAPM (Pre-2000):

	Alpha	Alpha_t	Alpha_p	R²	Alpha_Annual
Qnt 1	0.0045	2.8833	0.0045	0.8520	0.0537
Qnt 2	0.0034	4.4845	0.0	0.9497	0.0404
Qnt 3	0.0045	4.9799	0.0	0.9035	0.0542
Qnt 4	0.0043	3.8572	0.0002	0.8502	0.0511
Qnt 5	0.0082	4.6375	0.0	0.6613	0.0982

Equal-Weighted CAPM (Post-2000):

	Alpha	Alpha_t	Alpha_p	R²	Alpha_Annual
Qnt 1	0.0056	5.7224	0.0	0.8910	0.0676
Qnt 2	0.0057	6.1757	0.0	0.8838	0.0686
Qnt 3	0.0051	4.9947	0.0	0.8737	0.0617
Qnt 4	0.0046	2.9053	0.0042	0.7339	0.0554
Qnt 5	0.0064	3.6049	0.0004	0.7204	0.0768

Equal-Weighted CAPM (Pre-2000):

	Alpha	Alpha_t	Alpha_p	R ²	Alpha_Annual
Qnt 1	0.0045	2.8833	0.0045	0.8520	0.0537
Qnt 2	0.0034	4.4845	0.0	0.9497	0.0404
Qnt 3	0.0045	4.9799	0.0	0.9035	0.0542
Qnt 4	0.0043	3.8572	0.0002	0.8502	0.0511
Qnt 5	0.0082	4.6375	0.0	0.6613	0.0982

Equal-Weighted CAPM (Post-2000):

	Alpha	Alpha_t	Alpha_p	R ²	Alpha_Annual
Qnt 1	0.0056	5.7224	0.0	0.8910	0.0676
Qnt 2	0.0057	6.1757	0.0	0.8838	0.0686
Qnt 3	0.0051	4.9947	0.0	0.8737	0.0617
Qnt 4	0.0046	2.9053	0.0042	0.7339	0.0554
Qnt 5	0.0064	3.6049	0.0004	0.7204	0.0768

Table 5.1

Equal-Weighted FF5:

	Alpha	Alpha_t	Alpha_p	Beta_mkt	Beta_smb	Beta_hml	Beta_rmw	Beta_cma	R ²	Alpha_annual	Significant_alpha
Qnt 1	0.00618	9.30832	0.0	1.03692	-0.02405	-0.50178	0.13032	-0.04856	0.93545	0.07421	True
Qnt 2	0.00346	5.94738	0.0	1.03164	-0.00819	-0.08000	0.23145	0.13328	0.93296	0.04152	True
Qnt 3	0.00363	5.93682	0.0	1.00463	-0.04048	0.16658	0.07400	0.11708	0.91860	0.04353	True
Qnt 4	0.00223	3.61971	0.00034	1.01968	0.04110	0.49659	0.13174	0.08931	0.92061	0.02674	True
Qnt 5	0.00577	6.85369	0.0	1.00544	0.20832	0.72985	-0.07952	-0.01156	0.87171	0.06925	True

Value-Weighted FF5:

	Alpha	Alpha_t	Alpha_p	Beta_mkt	Beta_smb	Beta_hml	Beta_rmw	Beta_cma	R ²	Alpha_annual	Significant_alpha
Qnt 1	-0.00317	-2.82684	0.00503	1.03857	0.78776	-0.25347	-0.24700	-0.07699	0.89522	-0.03801	True
Qnt 2	-0.00176	-2.17614	0.03035	1.03786	0.76314	0.06155	0.08686	-0.02305	0.92411	-0.02117	True

Qnt 3	-0.0002 4	-0.3213 8	0.74815	0.96501	0.76508	0.30346	0.12642	0.00407	0.92112	-0.00289	False
Qnt 4	0.00110	1.56404	0.11890	0.90773	0.70105	0.35617	0.08082	0.16977	0.91873	0.01316	False
Qnt 5	0.00405	3.01096	0.00283	0.86944	0.85527	0.51059	-0.07955	0.22511	0.77875	0.04855	True

Table 5.2

Equal-Weighted 2×5 Double Sort Results (CAPM / FF3 / FF5):

Portfolio	CAPM Alpha	CAPM t	FF3 Alpha	FF3 t	FF5 Alpha	FF5 t	FF5 Alpha Annual	R² (FF5)
S0	-0.0086	-3.0061	-0.0057	-2.8943	-0.0046	-2.4061	-0.0548	0.7660
S1	-0.0029	-1.3314	-0.0012	-0.8402	-0.0010	-0.7182	-0.0126	0.7938
S2	0.0024	1.2550	0.0033	2.5521	0.0030	2.2535	0.0359	0.7735
S3	0.0035	1.9725	0.0040	3.3197	0.0038	3.0885	0.0459	0.7642
S4	0.0066	2.6808	0.0067	3.7448	0.0064	3.4902	0.0767	0.6647
B0	-0.0053	-3.2467	-0.0022	-2.9946	-0.0014	-1.8927	-0.0168	0.9504
B1	-0.0023	-2.2783	-0.0011	-1.7845	-0.0015	-2.3859	-0.0177	0.9495
B2	-0.0009	-0.9660	-0.0008	-1.2832	-0.0016	-2.7251	-0.0187	0.9497
B3	-0.0005	-0.4912	-0.0011	-1.9600	-0.0017	-3.2169	-0.0209	0.9500
B4	0.0001	0.1138	-0.0010	-1.6197	-0.0019	-3.0930	-0.0227	0.9378

Value-Weighted 2×5 Double Sort Results (CAPM / FF3 / FF5):

Portfolio	CAPM Alpha	CAPM t	FF3 Alpha	FF3 t	FF5 Alpha	FF5 t	FF5 Alpha Annual	R² (FF5)
S0	0.0237	6.1138	0.0292	12.5621	0.0318	15.4804	0.3815	0.8326
S1	0.0195	5.9545	0.0236	12.0502	0.0258	14.2379	0.3093	0.8238
S2	0.0174	7.5491	0.0194	15.1476	0.0200	15.6331	0.2396	0.8474
S3	0.0171	8.1806	0.0183	15.3429	0.0185	15.8290	0.2224	0.8493

S4	0.0247	9.5447	0.0251	15.6979	0.0247	15.3083	0.2966	0.7754
B0	0.0054	5.4131	0.0070	10.4908	0.0070	10.3402	0.0843	0.9329
B1	0.0039	6.2949	0.0039	6.4382	0.0032	5.4332	0.0380	0.9355
B2	0.0050	7.0716	0.0043	6.6836	0.0036	5.6201	0.0431	0.9101
B3	0.0043	5.5896	0.0030	5.4819	0.0026	4.6838	0.0308	0.9332
B4	0.0054	4.6910	0.0031	5.0607	0.0030	4.7688	0.0360	0.9213